

3.11, 12, 18, 19 5.

3-11

$$x = 3t - 4t^2 + t^3, m = 3kg$$

$$(1) v = 3 - 8t + 3t^2.$$

$$v_0 = 3 m/s, v_4 = 19 m/s$$

$$W = \Delta E_k = \frac{1}{2}mv_4^2 - \frac{1}{2}mv_0^2 = 528 J$$

$$(2) a = -8 + 6t$$

$$a_1 = -8 + 6 = -2 m/s^2.$$

$$F_1 = ma_1 = -6 N.$$

$$v_1 = -2 m/s$$

$$P = F_1 v_1 = 12 W.$$

3.17 (1) 当 $N = 0$ 时脱离, 由 $mg \cos \theta - N = m \frac{v^2}{R}$
~~知~~ 及 $mgR(1 - \cos \theta) = \frac{1}{2}mv^2$

$$\text{知 } \theta = \arccos \frac{2}{3}$$

$$(2) v^2 = 2gR(1 - \cos \theta) \Rightarrow v = \sqrt{\frac{2}{3}gR}$$

$$(3) h = R(\cos \theta + 1) = \frac{5}{3}R, x_0 = R \sin \theta = \frac{\sqrt{5}}{3}R$$

$$v = \sqrt{\frac{2}{3}gR}, v_x = v \cos \theta, v_y = v \sin \theta$$

$$y = h = v_y t + \frac{1}{2}gt^2$$

$$\Delta x = v_x t = 0.705 m.$$

$$x = x_0 + \Delta x \approx 1.450 m.$$

3.18

$$(1) W = \frac{(2+4) \times 10}{2} + \frac{(8-4) \times (-5)}{2} = 20 \text{ J}$$

$$(2) W = \frac{1}{2} m v^2 - \frac{1}{2} m v_0^2 \quad \text{由 } v = \sqrt{24} \text{ m/s}$$

3.19 由 $v_m = \sqrt{gr}$, $r = l - d$

$$\text{又 } h = d - (l - d) = 2d - l$$

$$mgh = \frac{1}{2} m v_m^2$$

$$\text{故 } d_{\text{min}} = 0.6l$$

$$\Rightarrow d \geq 0.6l.$$

7.5 设井底为原点

$$H = 10 \text{ m}, m_0 = 10 \text{ kg}; \text{ 且 } k = 0.2 \text{ kg/m}$$

$$m(y) = m_0 - ky = 10 - \frac{1}{5}y.$$

$$F = m(y)g = (10 - \frac{1}{5}y)g.$$

$$W = \int_0^{10} F dy = 90g.$$

$$g = 9.8 \text{ m/s}^2 \Rightarrow W = 882 \text{ J}$$